

The field of complex p-adic numbers (Lunch Seminar)

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02 Dec 2024

Based on chapter 3 sections 3 and 4 of [Kob96]. In this lecture we will continue talking about extensions of $\mathbb{Q}$ with the final goal of showing that $\overline{\mathbb{Q}_p}$ is not complete and then proving that its completion $\mathbb{C}_p$ is algebraically closed (and complete by definition)

1 Finite extensions of $\mathbb{Q}_p$

Reminder 1. Let $K/\mathbb{Q}_p$ be a degree n extension. $|\cdot|_p$ extends to a norm on K by $|\alpha|_p := |\mathrm{Nm}_{K/\mathbb{Q}_p}(\alpha)|^{1/n}$. Now we can also extend the p -adic valuation to K by

$$\mathrm{ord}(\alpha) = -\log_p(|\alpha|_p) = -\frac{1}{n} \log_p(|\mathrm{Nm}_{K/\mathbb{Q}_p}(\alpha)|_p) \in \frac{1}{n}\mathbb{Z}$$

The image of ord is a subgroup of $\frac{1}{n}\mathbb{Z}$ containing $\mathbb{Z}$, and so it is of the form $\frac{1}{e}\mathbb{Z}$ for some $1 \leq e \mid n$. This makes K into a discrete valuation field, with value group $\frac{1}{e}\mathbb{Z}$, valuation ring $R = \{\alpha \in K \mid |\alpha|_p \leq 1\}$, maximal ideal $\mathfrak{m} = \{\alpha \in K \mid |\alpha|_p < 1\}$

Definition 2.

1. We call e the index of ramification of $K/\mathbb{Q}_p$.
2. If $e = 1$ we say K is unramified and if $e = n$ we say K is totally ramified.
3. We denote the residual degree by $f := [R/\mathfrak{m} : \mathbb{F}_p]$, which is the degree of the residue field over $\mathbb{F}_p$.
4. A uniformizer is any element $\pi \in K$ with $\mathrm{ord}(\pi) = \frac{1}{e}$. Any uniformizer is a generator of $\mathfrak{m}$ and so any element can be written as $\pi^m \cdot u$ for some $m \in \mathbb{Z}$ and $u \in R^\times$ a unit.

Claim 3. $n = e \cdot f$

sketch of proof. Take some $\pi \in K$ uniformizer and $\overline{a_1}, \dots, \overline{a_f} \in R/\mathfrak{m}$ basis over $\mathbb{F}_p$. Now show that $\{\pi^i a_j\}_{i \leq e, j \leq f}$ is a basis of K over $\mathbb{Q}_p$. $\square$

We will now show that every finite extension of $\mathbb{Q}_p$ can be decomposed as an unramified extension followed by a totally ramified extension and even elaborate specifically how such extensions look.

Claim 4. If K is totally ramified and $\pi \in K$ is a uniformizer, then π satisfies an "Eisenstein equation":

$$\pi^e + a_{e-1}\pi^{e-1} + \dots + a_0 = 0$$

where $a_i \in \mathbb{Z}_p$ $a_i \equiv 0 \pmod p$ for all i , and $a_0 \not\equiv 0 \pmod{p^2}$. Conversely, if α is a root of such an Eisenstein equation over $\mathbb{Q}_p$, then $\mathbb{Q}_p(\alpha)$ is totally ramified over $\mathbb{Q}_p$ and of degree e .

Claim 5.

1. There is exactly one unramified extension K_f^{unram} of $\mathbb{Q}_p$ of degree f , which is obtained by adjoining a primitive $(p^f - 1)$ -th root of unity.

2. If K is an extension of $\mathbb{Q}_p$ of degree n , index of ramification e , and residual degree f , then $K = K_f^{\text{unram}}(\pi)$, where π satisfies an Eisenstein polynomial with coefficients in K_f^{unram} .

Corollary 6. If $K/\mathbb{Q}_p$ is a finite extension with index of ramification e , residual degree f , and $\pi \in K$ a uniformizer, then every $\alpha \in K$ can be written uniquely as $\sum_{i=m}^{\infty} a_i \pi^i$, where $m = e \cdot \text{ord}(\alpha)$ and $a_i^{p^f} = a_i$, $\forall i$.

Claim 7 (Krasner's Lemma). Let K be a complete non-archimedean field with characteristic 0 (e.g. $\mathbb{Q}_p$ or a finite extension of $\mathbb{Q}_p$), $a, b \in \overline{K}$ such that $|b - a| < |a' - a|$ for every a' Galois conjugate of a , then $K(a) \subseteq K(b)$.

Proof. Assume on the contrary that $a \notin K(b) =: L$, so $[L(a) : L] > 1$ is a non-trivial separable extension meaning there is an L -linear map $\sigma : \overline{K} \rightarrow \overline{K}$ such that $\sigma(a) \neq a$, otherwise it would have been trivial.

$$\begin{aligned} |b - a| &= |\sigma(b - a)| = |b - \sigma(a)| \\ \Rightarrow |a - \sigma(a)| &\leq \max(|b - a|, |b - \sigma(a)|) = |b - a| \end{aligned}$$

Which contradicts $|b - a| < |a' - a|$ for every conjugate a' . □

2 More on $\mathbb{C}_p$

Since $\overline{\mathbb{Q}_p}$ is a union of finite extensions, we can extend $|\cdot|_p$ to $\overline{\mathbb{Q}_p}$. Firstly, we note that $\overline{\mathbb{Q}_p}$ is not complete with regards to that norm.

Theorem 8. $\overline{\mathbb{Q}_p}$ is not complete.

The nicer proof is the following

rough sketch of non-constructive proof. Sketch:

1. Show that $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ is countably infinite dimensional.
2. Take $\alpha_1, \alpha_2, \alpha_3, \dots \in \overline{\mathbb{Q}_p}$ basis. So $\overline{\mathbb{Q}_p} = \bigcup_{n=1}^{\infty} \text{Span}(\alpha_1, \dots, \alpha_n)$ is a countable union of nowhere dense closed sets and so by (the contrapositive of) Baire category theorem $\overline{\mathbb{Q}_p}$ is not complete.
3. Another option is to use $\overline{\mathbb{Q}_p} = \bigcup_{n=1}^{\infty} \{x \mid [Q_p(x) : Q_p] \leq n\}$

□

But we will go through a more elementary constructive proof, for which we need the following claim:

Exercise 9. Let $\alpha \in \overline{\mathbb{Q}_p}$ be algebraic of degree n . Show that there exists N such that for every polynomial $P \in \mathbb{Z}_p[x]$ whose coefficients aren't all divisible by p and $\deg(P) < n$ we have $P(\alpha) \not\equiv 0 \pmod{p^N}$.

proof of theorem. Let's construct a sequence $(a_i)_{i=1}^{\infty}$ in $\overline{\mathbb{Q}_p}$ which is Cauchy and can't be convergent. For every $i \geq 0$ take b_i to be a primitive $p^{2^i} - 1$ -th root of unity. Note that for every $i > j$ b_j is a power of b_i , because $p^{2^j} - 1 \mid p^{2^i} - 1$. Now we define

$$a_i = \sum_{j=0}^i b_j p^{N_j} \in \mathbb{Q}_p(b_i)$$

Where we choose N_j inductively as follows: $N_0 = 0$ and given $N_0, \dots, N_i$, denote $K = \mathbb{Q}_p(b_i)$, which we proved is an unramified extension of degree 2^i , so $\mathbb{Q}_p(a_i) \subseteq K$ by definition of a_i . We see that $K \subseteq \mathbb{Q}_p(a_i)$ because if we assume not, then there is an automorphism $\sigma : K \rightarrow K$ such that $\sigma(a_i) \neq a_i$, but

$$\sum_{j=0}^i \sigma(b_j) p^{N_j} = \sigma(a_i) = a_i = \sum_{j=0}^i b_j p^{N_j}$$

contradicts the uniqueness of the expansion (note specifically that $\sigma(b_i) \neq b_i$). So a_i is algebraic of degree 2^i and by the exercise we can choose $N_{i+1} > N_i$ such that a_i does not satisfy any congruence of the form

$$P(a_i) \equiv 0 \pmod{p^{N_{i+1}}}$$

where $P \in \mathbb{Z}_p$, whose coefficients aren't all divisible by p and $\deg(P) < 2^i$.

This sequence is Cauchy because for every $i > j$ we have $p^{N_j} \mid a_i - a_j$ (equivalently $|a_i - a_j| < p^{-N_j}$)

This sequence is not convergent: Suppose $a \in \overline{\mathbb{Q}_p}$ is the limit of the sequence. a is algebraic so it satisfies some polynomial $P \in \mathbb{Z}_p[x]$. Choose some i such that $2^i > \deg(P)$. So we have $a \equiv a_i \pmod{p^{N_{i+1}}}$ and then

$$P(a_i) \equiv P(a) \equiv 0 \pmod{p^{N_{i+1}}}$$

a contradiction to the construction of a_i . □

Definition 10. Denote by $\mathbb{C}_p$ the completion of $\overline{\mathbb{Q}_p}$ with regards to $|\cdot|_p$.

Theorem 11. $\mathbb{C}_p$ is algebraically closed.

Proof. Take $P \in \mathbb{C}_p[x]$ monic irreducible (hence separable). The plan is to find $\beta \in \mathbb{C}_p$ such that Krasner's lemma applies for β and $\alpha \in \overline{\mathbb{C}_p}$ one of the roots of P , then we will get $\mathbb{C}_p(\alpha) \subseteq \mathbb{C}_p(\beta) = \mathbb{C}_p$, so P has a root in $\mathbb{C}_p$ and we finished. Write $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ for some $a_0, \dots, a_{n-1} \in \mathbb{C}_p$.

$\overline{\mathbb{Q}_p}$ is dense in $\mathbb{C}_p$, so we can choose $b_0, \dots, b_{n-1} \in \overline{\mathbb{Q}_p}$ such that $|a_i - b_i| < \delta$, $\forall i$ for some $0 < \delta < \frac{1}{n}$ sufficiently small we'll choose later.

Let β be a root of $Q(x) = x^n + b_{n-1}x^{n-1} + \dots + b_0$ (so $\beta \in \overline{\mathbb{Q}_p}$). Note that

$$|\beta|^n = \left| \sum_{i=0}^{n-1} b_i \beta^i \right| \leq \sum_{i=0}^{n-1} |b_i - a_i| |\beta|^i + \sum_{i=0}^{n-1} |a_i| |\beta|^i \leq \begin{cases} 1 + \sum_{i=0}^{n-1} |a_i| & \text{If } |\beta| \leq 1 \\ |\beta|^{n-1} (1 + \sum_{i=0}^{n-1} |a_i|) & \text{If } |\beta| > 1 \end{cases}$$

Either way we find that there is some $C > 0$ depending only on P such that $|\beta| < C$. Now we can see that

$$|P(\beta)| = |(P - Q)(\beta)| \leq \sum_{i=0}^{n-1} |a_i - b_i| |\beta|^i \leq \delta \sum_{i=0}^{n-1} C^i$$

Denote $\lambda := \min_{i \neq j} |\alpha_i - \alpha_j|$ where $\alpha_1, \dots, \alpha_n$ are the roots of P in $\overline{\mathbb{C}_p}$. choose delta small enough such that

$|P(\beta)| < (\frac{\lambda}{3})^n$ and we get

$$\prod_{i=1}^n |\beta - \alpha_i| = |P(\beta)| \leq \left(\frac{\lambda}{3}\right)^n$$

so at least one of the terms on the left hand side is $< \frac{\lambda}{3}$, denote $\alpha = \alpha_i$ that root and we get for every $j \neq i$

$$|\beta - \alpha| \leq \frac{\lambda}{3} < \lambda \leq |\alpha - \alpha_j|$$

So we finished. □

Remark. In the proof above we actually used Krasner's lemma to show that the operations of taking p -adic completion and taking algebraic closure commute. Looking again at the proof of Krasner's lemma we see it is true for every complete non-archimedean field and so the proof above reveals a general phenomenon.

Fun fact: $\mathbb{C}_p$ and $\mathbb{C}$ are isomorphic as rings by abstract algebraic consideration (talking about transcendence bases and such), but are completely different as topological rings.

References

- [Kob96] N. Koblitz. *p-adic Numbers, p-adic Analysis, and Zeta-Functions*. Graduate Texts in Mathematics. Springer New York, 1996. ISBN: 9780387960173. URL: https://perso.univ-rennes1.fr/serge.cantat/Documents/Koblitz_ENS_2020.pdf.